

CS4243 Cheatsheet

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Misc. Operations

Mean/Std. dev. $\bar{x} = \frac{1}{n} \sum_i x_i$, $\sigma = \sqrt{\frac{1}{n} \sum_i (x_i - \bar{x})^2}$

Normalisation $x' = \frac{x - \mu}{\sigma}$ (or min-max: $\frac{x - \min}{\max - \min}$)

Gaussian (1D) $G(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2}$

Gaussian (2D) $G(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}$

Accuracy: $\frac{TP + TN}{TP + TN + FP + FN}$ **Precision:** $\frac{TP}{TP + FP}$

Recall: $\frac{TP + FN}{TP + FN}$ **F1-score:** $\frac{2PR}{P + R}$

Line eqns: $y = mx + b$; $x \cos \theta + y \sin \theta = \rho$ **Euclidean dist:** $d = \sqrt{\sum_i (x_i - y_i)^2}$

Orientation: $\theta = \tan^{-1}\left(\frac{f_y}{f_x}\right)$ **Magnitude:** $\|\nabla f\| = \sqrt{f_x^2 + f_y^2}$
 $\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$ $\|\nabla f\| = \sqrt{G_x^2 + G_y^2}$

Lecture 1 - Computer Vision

Digital Image Acquisition

Illuminance ($i(x, y)$, $0 \leq i \leq \infty$): amount of source illumination incident on scene.

Reflectance ($r(x, y)$, $0 \leq r \leq 1$): amount of source illumination reflected by objects on scene.

Exposure: amount of time incident light can reach imaging sensor.

Aperture: how much light you let in (too much: bokeh, too narrow sharp image)

Shutter Speed: how much time you give light to accumulate (slow: motion blur, fast: freeze action)

	Spatial Resolution	Intensity Resolution
Definition	smallest discernible detail	smallest discernible δ intensity level
Measurement	pixels / unit distance	bit depth / pixel value
Determined by	sampling	quantisation
Effects of low res	pixellation	false contouring

File size (bits): $b = M \cdot N \cdot k$

M : rows, N : columns, k : bits/pixel, b : total bits

File Extensions

.bmp: raw bit file (largest size)

.png: predicts pixels based on neighbours, reconstructs original pixel value (lossless)

.jpg: 8x8 image blocks represented by 64 Discrete Cosine Transform coefficients, reconstruction is approximate (lossy). Zigzag ordering captures larger coefficients first; more efficient for Huffman entropy encoding

Colour Representation

RGB Space $im(x, y, b) = y$ pixels down, x pixels to right in the b th channel

Convert to grayscale: weighted sum of each colour

HSV Space: Hue = colour (0–360°, 0 = red (orange → pink); Saturation = purity (0–1, 0 = grey); Value = brightness (0–1, 0 = black). Separates colour from intensity → robust to lighting

Colour & intensity are complementary; intensity is often stronger information for shape and structure

Bayer Filter

Each pixel only stores one color sample over it (RGB). Reduces intensity resolution.

$$g(x, y) \begin{cases} [R, G, B]_B = [f(x+1, y+1), f(x+1, y), f(x, y)] \\ [R, G, B]_{GB} = [f(x, y+1), f(x, y), f(x-1, y)] \\ [R, G, B]_{GR} = [f(x+1, y), f(x, y), f(x, y-1)] \\ [R, G, B]_R = [f(x, y), f(x-1, y), f(x-1, y-1)] \end{cases}$$



Lecture 2 - Point Processing and Filtering

Point processing: $x_{ij} = f(p_{ij})$

Filtering: $x_{ij} = f(w_{ij})$

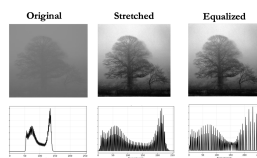
Point Processing

Brightness = $x_{ij} = p_{ij} + b$ (clip when needed)

Intensity scaling = $x_{ij} = a \cdot p_{ij}$

Image Standardisation Preprocess to remove arbitrary brightness/contrast baselines. Algorithms process better when inputs are centered around 0.

Histogram Stretching vs. Equalisation



Stretching: linear rescale to full range

$$s = \frac{r - r_{\min}}{r_{\max} - r_{\min}}(L - 1)$$

Pros: improve contrast; keeps relative brightness

Cons: limited if histogram spans most of existing range

Equalisation: redistribute via CDF for uniform histogram

$$s = (L - 1) \sum_{k=0}^r p(k), \quad s_{\text{output}} = k \cdot CDF(s_{\text{input}})$$

Pros: more dramatic contrast change; reveals poorly represented ranges automatically

Cons: can amplify noise; context-blind

Cross-Correlation and Convolution

Purpose: smoothing/averaging with neighbours reduces impulse (salt-and-pepper) noise; also used for edge detection and template matching.

Cross-Correlation: slide kernel F as-is, multiply & sum

$$x_{ij} = \sum_u \sum_v f_{uv} \cdot p_{i+u, j+v}, \quad X = P \circledast F$$

Convolution: flip kernel 180° first, then multiply & sum. Commutative and associative.

$$x_{ij} = \sum_u \sum_v f_{uv} \cdot p_{i-u, j-v}, \quad X = P * F$$

Equal when: kernel is symmetric under 180° rotation, i.e. $F = F_{\text{flip}}$ (e.g. box filter, Gaussian).

Gaussian Kernel

Used as the filter f_{uv} in cross-correlation/convolution above to smooth/blur. Larger σ = more blur.

Detail extraction (unsharp masking):

Details = Orig. – Blurred $\Rightarrow$ Sharpened = Orig. + α · Details

Lecture 3 - Gradients

Sobel Filter

Horizontal Sobel filter: $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$ $\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$

Vertical Sobel filter: $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ $\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$

Vertical filter S_y detects horizontal edges; horizontal filter S_x detects vertical edges (edge $\perp$ gradient).

Gradient points from dark to bright intensities. Edge occurs at peaks.

2D Separable filter: $K = uv^T$ (outer product of column u , row v). Conv with K = conv with u then v . Cost: $2n$ vs n^2 mults.

Example — Vertical Sobel filter:

- Smooth horizontally: $[1, 2, 1]$
- Differentiate vertically: $[1, 0, -1]^T$

$$S_y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

Computing gradients

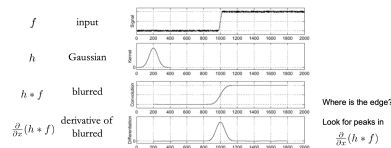
- Convolve gradient filter with image f to compute image gradient components

$$G_x = \frac{\partial f}{\partial x} = S_x * f \quad G_y = \frac{\partial f}{\partial y} = S_y * f$$

- Compute gradient orientation and magnitude

Gaussian Derivative and Laplace Filters

Differentiation accentuates noise (multiplicative factor proportional to frequency) $\Rightarrow$ blur before using derivative features to remove noise (but the edge will blur).



Stronger Gaussian returns more high-frequency noise $\rightarrow$ spreads edge across larger region.

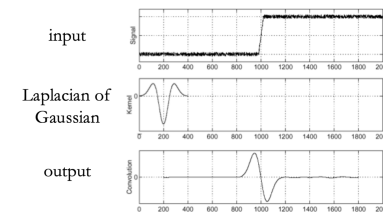
Differentiation Property: $\frac{\partial}{\partial x}(h * f) = \left(\frac{\partial}{\partial x}h\right) * f$

Differentiation is a convolution operation: $f * [1 \ 0 \ -1]$ (x-direction) or $f * [1 \ 0 \ -1]^T$ (y-direction)

Laplacian Operator

Continuous: $\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$

Discrete: $\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$



Zero crossings. Localise edges more accurately but less convenient to use.

Canny Edge Detector

Deriv. of Gaussian $\rightarrow$ Grad. mag/orient. $\rightarrow$ NMS $\rightarrow$ Hysteresis thresh.

Non-Maximum Suppression: Only keep local maxima for edges (so edges will be one pixel wide. *Algorithm* Given magnitude map $M(x, y)$ and orientation map $\theta(x, y)$, for every (x, y)

- Find two neighbours in that direction according to the nearest angle $(+\theta, -\theta)$
- Interpolate the two neighbour values, with value $= x * (a - b) + b$, where a, b are the two real neighbour pixel magnitudes, and x is the fractional distance, $x = \tan(\theta \bmod 90)$



- Compare centre pixel to interpolated neighbours.

If $M(x, y) \geq$ both neighbours: keep $M(x, y)$ Else: suppress and set $M(x, y) = 0$

Hysteresis Thresholding: Set two thresholds, H to start edge curves, and L to grow and link edges (if connected to boundary, mark said pixel as boundary) *Algorithm* E_H : all values passing H , E_L : all values passing L . Let $D = E_L - E_H$ (keep only edges of D connected to E_H)

Final image: $E_H \cup D'$

Hough Transform Algorithm Given magnitude map $M(x, y)$ and orientation map $\theta(x, y)$, for every (x, y)

- Quantise parameter space (m, b) by picking a range and step size, then discretise.
- Initialise accumulator array $A(m, b)$. Set each $A(m, b) = 0, \forall m, b$.
- $\forall (x_i, y_i), \forall m : b = x_i m + y_i$. Increment $A(m, b) = A(m, b) + 1$
- Threshold and find local maxima of $A(m, b)$. $y = m^* x + b^*$

Use $x \cos \theta + y \sin \theta = \rho$ — finite parameters ($\theta \in [0, 360]$) vs. unbounded gradient m

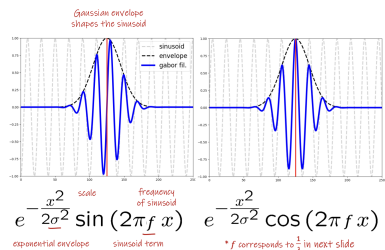
Can be made more efficient by voting only near its measured gradient orientation.

Lecture 4 - Regions

Filter banks: collection of multiple filter kernels, where one filter kernel responds to one local pattern. **Gabor**

Laplacian of Gaussian Filter

Filter: fix general math function with a range of parameters. Composed of sinuisoids and Gaussian envelope.



f : frequency selects spacing between waves; σ : scale determines size of region covered by Gaussian envelope.
2D Gabor formula:

$$f_{mn} = \underbrace{\frac{1}{2\pi\sigma^2} \exp\left[-\frac{m^2 + \gamma^2 n^2}{2\sigma^2}\right]}_{\text{exponential (Gaussian) envelope selects a local area}} \underbrace{\sin\left[\frac{2\pi(\cos[\omega]m + \sin[\omega]n)}{\lambda} + \phi\right]}_{\text{2D-Sinusoid selects orientation and frequency}}$$

ω – orientation in 2D; λ – wavelength of sinusoid (inversely proportional to f); ϕ – phase of sinusoid; shift from center of kernel

γ – aspect ratio of envelope coverage ($\gamma = 1$ means circular, else elliptical); σ – rate of decay of exp. envelope

Multi-Scale Reasoning

Smoothing: suppress fine detail $\rightarrow$ focus on larger structure.

Gaussian pyramid: blur, then downsample (fixed rate). Smooth first to avoid aliasing (high freq. masquerading as low).

Scale selection: filter at multiple scales; pick the response peak (filter size matching structure size).

Single vs Multi-scale Representation Given r_i, ρ_i corresponding to response and scale at i .

Single scale: $r = [r_1, r_2, \dots, r_k]$

Multi-scale: $r = [r_1^{\rho_1}, r_2^{\rho_1}, \dots, r_k^{\rho_1}, r_1^{\rho_2}, \dots, r_k^{\rho_2}]$.

Textures

Textures occur over an area. To detect, group pixels by region Ω and average filter responses over region.

- $r(x, y) = [r_1(x, y), r_2(x, y), \dots, r_k(x, y)]$ Run k different filters to get k responses.
- $t_k(\Omega) = \frac{1}{|\Omega|} \sum |r_k(x, y)|$. Take absolute value of every pixel's response to filter k in region Ω and sum. Divide pixels by $|\Omega|$ for average.
 $t_k(\Omega)$ is a pooled response of filter k across Ω
- $t(\Omega) = [t_1(\Omega), t_2(\Omega), \dots, t_k(\Omega)]^T$

To compare textures, $D(\Omega_A, \Omega_B) = ||t(\Omega_a) - t(\Omega_a)||$, i.e. D is the Euclidean distance.

Classifiers

- Nearest Neighbour Classifier: $i^* = \arg \min_i ||t(\Omega) - t(\Omega_i)||$ where $\hat{c} = y_{i^*}$

- Nearest Class Mean: $\mu_c = \frac{1}{N_i} \sum i : y_i = ct(\Omega_c)$ where $\hat{c}(\Omega) = \arg \min ||t(\Omega) - \mu_c||$



Topic 5 - CNNs

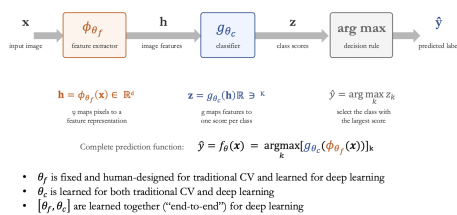
Receptive field: set of image pixels an intermediate output pixel depends on.

Recursive formula (per layer l , with kernel k_l , stride s_l):
 $r_l = r_{l-1} + (k_l - 1)j_{l-1}$, $j_l = j_{l-1}s_l$ base case:
 $r_0 = 1, j_0 = 1$

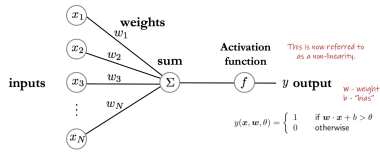
Training: Training Images $\rightarrow$ Image Features $\rightarrow$ Training with Labels $\rightarrow$ Learned Classifier

Testing: Test Image $\rightarrow$ Image Features $\rightarrow$ Learned Classifier $\rightarrow$ Prediction

Stages of Classifier



Perceptron



No of learnable parameters = weights (edges) + perceptrons (neurons).

Neurons are total of all possible output nodes (all layers except input). We add perceptrons as bias terms.

MLP Formalism

Input: $\mathbf{a}^{(0)} = \mathbf{u}$; $\mathbf{u}$: input feature vector

Hidden layers:

$$\mathbf{a}^{(\ell)} = \sigma(W^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}), \ell = 1, \dots, L-1$$

$\mathbf{a}^{(\ell)}$: representation in layer ℓ .

$W^{(\ell)}$: weights in layer ℓ .

$\mathbf{b}^{(\ell)}$: biases in layer ℓ .

σ : non-linear activation function

Output layer: $\mathbf{z} = \mathbf{a}^{(L)} = W^{(L)}\mathbf{a}^{(L-1)} + \mathbf{b}^{(L)}$

Convolution

No of learnable parameters:

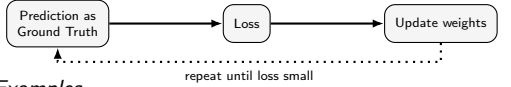
$$k^2 \cdot C_{in} \cdot C_{out} + C_{out}$$

Strided. Downsample each feature map. Reduce spatial resolution. **Dilated.** Enlarge receptive field without adding params or reducing resolution.

$$H_{output} = \left\lfloor \frac{H_{input} + 2P - D(K-1) - 1}{S} \right\rfloor + 1$$

Logits: unrestricted real-value scales $\rightarrow$ convert into probability distribution using softmax.

Loss functions



Examples

- Localisation loss: $L_{loc} = (\hat{x} - x)^2 + (\hat{y} - y)^2$. Distance-based, naturally continuous.
- $L_{0/1}(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y \text{ (correct)} \\ 1, & \hat{y} \neq y \text{ (wrong)} \end{cases}$. Discrete
- Cross-entropy loss: $L_{CE} = -\log(p_y)$. Continuous — softly penalizes low confidence in the correct class

Object Detection

Window-based: Build/train model $\rightarrow$ Generate candidate windows $\rightarrow$ Score candidates $\rightarrow$ Resolve detections with NMS.

Performance Evaluation Area of Overlap Measure

$$AO(B_{gt}, B_p) = \frac{|B_{gt} \cap B_p|}{|B_{gt} \cup B_p|}$$

mAP (mean Average Precision): standard object-detection metric, 0–1 or 0–100%. We want high precision (few false positives) and high recall (few false negatives).

- $AP(c, \tau)$ = area under precision–recall curve (class c , $\text{IoU} \geq \tau$)
- $\text{mAP}(\tau)$ = mean AP across classes

$$AP(c, \tau) = \int_0^1 p(r) dr \quad \text{mAP}(\tau) = \frac{1}{|C|} \sum_{c \in C} AP(c, \tau)$$

CNN-Based Object Detection

R-CNNs (Region-Based ConvNets): Generate region candidates, extract CNN features for each region, regress bounding box (x, y, h, w) for each region, predict label per region.

Fast R-CNNs: Extract CNN features in one pass by locating (x, y) in CNN feature maps and extracting features w ROI pooling.

YOLO (v2) – Anchor-based Detection

- Split image into $S \times S$ cells, each with predefined **anchors** (boxes w/ fixed sizes & aspect ratios)
- Predict at every (grid location, anchor) pair
- Objectness:** $P(\text{obj})$ for box. Predict class + box, skip if no object.
- Class score:** $P(\text{class} | \text{obj})$
- Box adjustment:** $P(\text{coords} | \text{obj})$, offsets to anchor
- Loss:** $\mathcal{L} = \mathcal{L}_{\text{obj}} + \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{class}}$; $\mathcal{L}_{\text{obj}}$: all anchors; $\mathcal{L}_{\text{box}}, \mathcal{L}_{\text{class}}$: object anchors only

Topic 6 - Segmentation

Separation of images into regions/objects.

Semantic: each pixel has a class label. Identify foreground and background.

Instance: Each pixel has an object it belongs to.

Panoptic: Combine semantic and instance.

k-Means Clustering: Iteratively re-assign points to nearest cluster center

- Randomly initialize the cluster centers $c_1, \dots, c_K$
- For each point p_i , find the closest c_j to put p_i in
- Set c_j to be mean of points in cluster j
- Repeat, if c_j have changed up to some threshold

- Pros: Simple, Converges to local min.
- Cons: Setting K , Sensitive to initial centers (Since k-means converges to local min.), Sensitive to outliers (Can add more clusters), Assumes spherical clusters

Simple Linear Iterative Clustering (SLIC)

- Superpixel:** Group of pixels sharing common traits.
- No of pixels: n_{tp} ; Target no of superpixels: n_{sp}
- Initial width of each superpixel: $s = \sqrt{n_{tp}/n_{sp}}$
- Features: $z = [r, g, b, x, y]$
- Color distance: $d_c = ||\langle r_j, g_j, b_j \rangle - \langle r_i, g_i, b_i \rangle||$
- Spatial distance: $d_s = ||\langle x_j, y_j \rangle - \langle x_i, y_i \rangle||$
- d_{cm} and $d_{sm} = s$ set as max. expected values of d_c and d_s respectively

$$D = \sqrt{\left(\frac{d_c}{d_{cm}}\right)^2 + \left(\frac{d_s}{d_{sm}}\right)^2} = \sqrt{d_c^2 + \left(\frac{d_s}{s}\right)^2} c^2$$

- Split img. into grid of size $s \times s$. Set cluster centers as lowest gradient position in 3×3 neighborhood from superpixel center.
- For each center, check distance to all pixels within $2s \times 2s$ neighborhood. Assign pixels to closest checked center.
- Update centers using mean; repeat until converged
- Optional: Replace superpixel region by average value to create stained glass effect

- Modification of k-Means: Not random initialization, Compute pixel's distance only to closest set of centers

Mean-Shift Clustering: Find local density maxima in feature space

- Attraction basin:** Region where all trajectories of centroids lead to same mode

- Cluster:** All data points in attraction basin of a mode

- For each data point:
 - Define window around and get centroid
 - Shift window to centroid
 - Repeat until window centroid stops moving
- Segment: run mean shift, merge pixels within basin
- Larger window size $\rightarrow$ Fewer clusters
- Pros: No assumptions on cluster shape, 1 parameter, Finds variable num. of modes (vs. specified k in k-Means), Robust to outliers
- Cons: Choosing h , scales poorly with feature space dimension